

Requirements Table — Digital Campus Library Reservation Gateway

Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #08 — Campus & Academic Operations

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow members to search the ISBN repository and place hold requests on currently loaned books to enter a FIFO waitlist.	High	Pass: User is assigned correct position in queue. Fail: A user already holding the maximum allowed books is able to place an additional reservation.	Core reservation functionality; guarantees fair, first-come-first-served access to high-demand titles.
FR-002	Functional	The system shall notify a student member that their reserved book is available for pickup within 5 minutes of the book being checked in and marked available.	High	Pass: The member receives a notification within 5 minutes of the book's check-in timestamp for 95% of test cases. Fail: Notification is delayed beyond 5 minutes or not delivered.	Prevents queue stagnation and wasted holding-shelf space by ensuring the member acts on the hold promptly. (Delivery channel and mechanism are left to the design phase.)
FR-003	Functional	The system shall automatically calculate overdue fines at a rate of ₹5 per day per book, capped at ₹200 per book, starting the day after the due date.	Medium	Pass: Calculated fine matches the formula exactly for test cases at 1, 10, 40, and 60+ days overdue (cap applied). Fail: Any test case returns an incorrect fine amount.	Ensures consistent, transparent, and auditable fine tracking without manual librarian calculation.
FR-004	Functional	The system shall allow a student member to cancel their own pending hold request at any time before the reserved copy is checked out to them.	Medium	Pass: Cancelled hold is removed from the queue immediately and the next member in line advances one position. Fail: Cancelled hold remains visible in the active queue.	Keeps the FIFO queue accurate and frees reserved copies for other waiting members as early as possible.
FR-005	Functional	The system shall allow the Head Librarian to manually reorder or remove entries in a book's holding queue for exceptional cases (e.g., a lost or damaged copy).	Low	Pass: The queue reflects the librarian's manual change and logs the librarian ID, timestamp, and reason. Fail: The change is applied without being logged, or is not applied at all.	Provides administrative flexibility to handle real-world exceptions that a purely automated FIFO queue cannot resolve.
NFR-001	Nonfunctional (Performance & Security)	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load. Fail: p95 latency exceeds 150 ms or security checks fail under load.	Search is the entry point to every workflow; slow or insecure search degrades the experience for the entire student body.
NFR-002	Nonfunctional (Reliability & Availability)	The system shall maintain at least 99.5% uptime during official library operating hours (8:00 AM–10:00 PM), excluding pre-announced maintenance windows.	High	Pass: Monitoring logs show uptime $\geq 99.5\%$ over a rolling 30-day period. Fail: Uptime falls below 99.5% during operating hours in any 30-day window.	Students depend on the gateway during open hours for time-sensitive holds and fine payments; downtime directly blocks core library operations.

Use-Case Flow Specification

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Use Case: UC-04 — Pay Overdue Fine

Primary Actor: Student Member

Supporting Actor: Payment Gateway (external); Head Librarian may optionally intervene via the «extend» relationship (UC-06 Apply Fine Waiver)

Preconditions

- Student has at least one book returned past its due date with an outstanding calculated fine greater than ₹0.

Postconditions

Success: The fine balance is reduced to ₹0 and a receipt is issued to the student.

Failure: The fine balance remains unchanged and no payment is recorded.

Main Success Scenario

1. Student Member opens “My Account” and views the outstanding fine balance.
2. System displays itemized fines per book (days overdue × rate, capped at ₹200).
3. Student selects “Pay Fine” and chooses a payment method (card / UPI / campus wallet).
4. System connects to the Payment Gateway and submits the transaction.
5. Payment is authorized.
6. System marks the fine as paid, updates the student's account balance to ₹0, and generates a receipt.
7. System emails the receipt to the student.
8. Use case ends successfully.

Alternate Flow

5a. Payment Declined

- 5a1. System displays a “Payment failed” message.
- 5a2. Student is prompted to select another payment method.
- 5a3. If payment fails twice more, the transaction is cancelled and the fine remains outstanding; student is notified to retry later. (This is also where UC-06 Apply Fine Waiver can optionally intervene, per the «extend» relationship in the use-case diagram.)

Flow Diagram

UC-04: Pay Overdue Fine